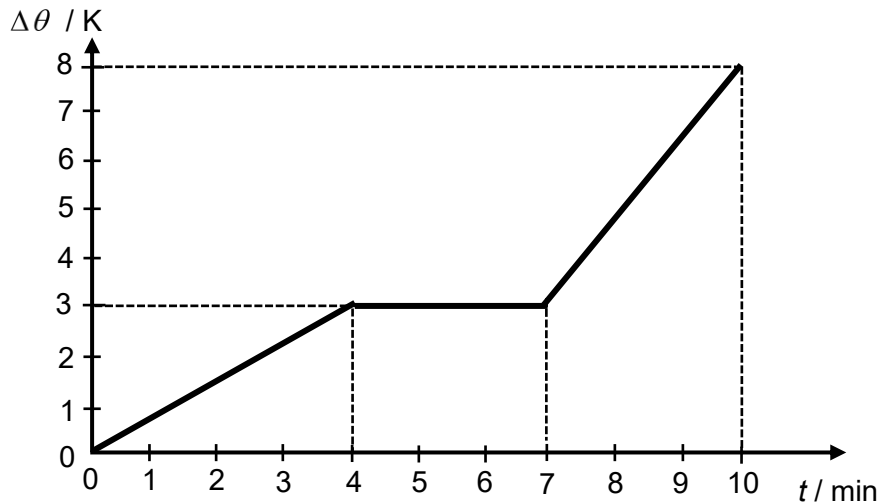


- 13 The graph shows the variation with time t of temperature change $\Delta\theta$ for 1 kg of a substance, initially solid at room temperature. The substance receives heat at a uniform rate of 2000 J min^{-1} .



What can be deduced from this graph?

- | | |
|----------|---|
| A | The specific heat capacity of the substance is greater when liquid than when solid. |
| B | The specific latent heat of fusion of the substance is 6000 J kg^{-1} . |
| C | The substance melts at a temperature of 3 K. |
| D | After 10 min, the substance is all gaseous. |

Answer: B

Option A :

$$Q = Pt = mc\Delta\theta$$

[Turn over

$$P = \frac{mc\Delta\theta}{t}$$

From the graph, $\frac{\Delta\theta}{t}$ is bigger for liquid (7 – 10 min) than solid (1 – 4 min).

Since P and m are constant, therefore specific heat capacity c is smaller when liquid than when solid.

Option B:

$$Q = mL$$

$$2000 \times 3 = 1 \times L$$

$$L = 6000 \text{ J kg}^{-1}$$

Option C:

The substance melts after an increase in temperature of 3 K from room temperature. The melting temperature is not 3 K.

Option D:

Unless the graph becomes horizontal again, we are unable to determine when the substance starts to become gaseous.

14 A small pendulum bob is displaced to one side and released from rest at time $t = 0$. The bob